CBET Memorization Sheet

Print this and keep it beside you. The CBET exam rewards recall of normal values, safety limits, device settings, units, standards, and formulas across every kind of equipment in a hospital. Every number on this sheet is fair game.

01 The exam itself

FACT	VALUE
Certifying body	**AAMI** Credentials Institute (**ACI**)
Questions	165 multiple choice; 150 scored, 15 unscored pretest items mixed in
Time and rules	3 hours, closed book, simple calculator allowed, no phones
Passing	Criterion-referenced cut score (modified Angoff), about 116 of 165 correct, roughly 70%
Healthcare Technology and Function	30%
Healthcare Technology Problem Solving	30%
Healthcare Information Technology	17%
Public Safety in the Healthcare Facility	10%
Anatomy and Physiology	7%
Fundamentals of Electricity and Electronics	6%

> Function and problem solving are 60% of the exam. For every device: what it measures or delivers, its settings and normal values, its alarms, its common faults, and the first check.

02 Anatomy and physiology

VITAL SIGN (ADULT, RESTING)	NORMAL RANGE	MEASURED BY
Heart rate	60 to 100 bpm (below 60 bradycardia, above 100 tachycardia)	**ECG**, pulse oximeter, **NIBP**
Respiratory rate	12 to 20 breaths per minute	Impedance pneumography, capnography
Blood pressure	About 120/80 **mmHg**; **MAP** = diastolic + (systolic minus diastolic) / 3, about 70 to 100	**NIBP**, arterial line
Oxygen saturation (SpO2)	95 to 100%	Pulse oximeter
End-tidal CO2 (EtCO2)	35 to 45 **mmHg**	Capnograph
Body temperature	37 C (98.6 F); fever above about 38 C	Thermistor, thermocouple, infrared
Fetal heart rate	110 to 160 bpm	Doppler ultrasound
Tidal volume	About 500 **mL** at rest (6 to 8 **mL**/kg ideal body weight on a ventilator)	Ventilator, spirometer
Cardiac output	About 5 L/min = heart rate x stroke volume (about 70 **mL**)	Derived

SYSTEM	KEY PARTS	WHAT THE TECHNICIAN'S DEVICES DO WITH IT
Circulatory	Heart (right atrium, right ventricle, left atrium, left ventricle), tricuspid, pulmonary, mitral, aortic valves, arteries, veins, capillaries	**ECG** records electrical activity; **NIBP** and arterial lines measure pressure; defibrillators and pacemakers control rhythm; balloon pumps assist

SYSTEM	KEY PARTS	WHAT THE TECHNICIAN'S DEVICES DO WITH IT
Blood flow	Body, vena cava, **RA**, tricuspid, **RV**, pulmonary valve, pulmonary artery, lungs, pulmonary veins, **LA**, mitral, **LV**, aortic valve, aorta, body	Right side pumps to lungs, left side to body
Conduction path	**SA** node, **AV** node, bundle of His, bundle branches, Purkinje fibers	P wave atrial depolarization, **QRS** ventricular depolarization, T wave ventricular repolarization
Respiratory	Trachea, bronchi, bronchioles, alveoli, diaphragm	Ventilators move gas; oximeters measure oxygen carried; capnographs measure **CO2** exhaled; spirometers measure volumes
Nervous	Brain (cerebrum, cerebellum, brainstem), spinal cord, peripheral and autonomic nerves	**EEG** records brain activity; **EMG** records muscle; stimulators test conduction; sympathetic raises heart rate, parasympathetic lowers it
Musculoskeletal	206 bones, joints, skeletal (voluntary), smooth (involuntary), and cardiac muscle	Physical therapy stimulators, traction, continuous passive motion
Gastrointestinal	Esophagus, stomach, small intestine (absorption), large intestine, liver, pancreas, gallbladder	Feeding pumps deliver enteral nutrition; suction removes; endoscopes view
Endocrine	Pancreas (insulin, glucagon), thyroid, adrenals (epinephrine), pituitary	Glucose meters, insulin pumps
Renal	Kidneys (nephrons filter blood), ureters, bladder	Dialysis replaces filtration
Skin	Epidermis, dermis, subcutaneous layer; barrier, temperature control, sensation	Electrode contact and skin impedance, burns, pressure injuries
Blood	Plasma, red cells (hemoglobin carries oxygen), white cells (immunity), platelets (clotting)	Analyzers count and measure; oximetry depends on hemoglobin

Directional terms and word parts

- Anterior front, posterior back; superior above, inferior below; medial toward the midline, lateral away; proximal near the trunk, distal far; supine face up, prone face down; planes: sagittal (left and right), coronal (front and back), transverse (top and bottom)
- Systole contraction, diastole relaxation. Hypertension high pressure, hypotension low. Hypoxia low oxygen, hypercapnia high **CO2**, apnea no breathing
- Prefixes: brady- slow, tachy- fast, hyper- high, hypo- low, a-/an- without, dys- difficult, peri- around, endo- inside
- Roots: cardio heart, pulmo/pneumo lung, nephro/reno kidney, hepat liver, neuro nerve, gastro stomach, derm skin, osteo bone, hemo/hemat blood, myo muscle, encephal brain
- Suffixes: -itis inflammation, -ectomy removal, -ostomy opening, -oscopy viewing, -graphy recording, -gram the record, -algia pain, -emia blood condition

ECG timing (normal adult)

- **PR** interval 0.12 to 0.20 s; **QRS** under 0.12 s; **QT** under about 0.44 s
- Paper speed 25 mm/s; 1 **mV** = 10 mm; small box 0.04 s (1 mm), large box 0.20 s (5 mm)
- Heart rate = 1500 / small boxes between R waves, or 300 / large boxes
- Ventricular fibrillation: chaotic, no **QRS**, shockable. Asystole: flat line, not shockable. Atrial fibrillation: irregular, no P waves

ANATOMY DETAIL	FACTS
Skin layers	Epidermis (stratum basale at its base) over dermis over hypodermis (deepest); sebaceous glands oil, sweat glands cool, vitamin D made in skin
Muscle types	Skeletal striated voluntary; cardiac striated involuntary; smooth non-striated involuntary (organs, vessels, uterus, bladder)
Skeleton	206 bones: 80 axial (skull, 33 vertebrae, ribs, sternum with manubrium and xiphoid, ossicles, hyoid) and 126 appendicular (scapula, clavicle, pelvis, humerus, radius, ulna, femur, patella, tibia, fibula)

ANATOMY DETAIL	FACTS
Brain	Forebrain: cerebrum (frontal plans, parietal senses, occipital sees, temporal hears), thalamus relays, hypothalamus drives the pituitary. Midbrain relays vision and hearing. Hindbrain: cerebellum coordinates, pons relays, medulla runs breathing and heart rate. Brain stem = midbrain + pons + medulla
Nerves	Somatic voluntary; autonomic involuntary (sympathetic fight or flight, parasympathetic rest and digest); neurons signal by action potentials across synapses
Digestion in order	Mouth (amylase, lipase), esophagus (epiglottis, lower esophageal sphincter), stomach (**pH** 1.5 to 3.5, pyloric sphincter releases chyme), duodenum, jejunum, ileum, large intestine (reabsorbs water); liver makes bile, glycogen, urea; gallbladder stores bile; pancreas enzymes and insulin
Lymph and immunity	Lymph filtered in nodes (neck, armpits, groin); tonsils, thymus (T cells), spleen (cleans blood), Peyer's patches; the immune system avoids, detects, and eliminates infection
Endocrine	Hypothalamus and pituitary control; thyroid metabolism; parathyroid (posterior thyroid lobes) raises calcium; adrenal cortex cortisol, medulla epinephrine; islets insulin and glucagon; pineal rhythms; negative feedback stabilizes, positive amplifies; the pons secretes nothing
Kidney	Cortex, medulla, pelvis; nephron: glomerulus in Bowman's capsule (blood pressure drives filtration), tubules reabsorb and secrete; ureter, bladder, urethra
Reproduction	Testes, epididymis, vas deferens, prostate; **FSH** sperm, **LH** testosterone. Ovaries (estrogen, progesterone), fallopian tube (fertilization), uterus (implantation)
Blood terms	Hemoglobin carries oxygen; hematocrit is the red-cell fraction (about 45%); about 5 liters total; lymphocytes, neutrophils, monocytes, eosinophils, basophils

03 Electricity and electronics

FORMULA	STATEMENT
Ohm's law	V = I x R; I = V / R; R = V / I
Power	P = V x I = I squared x R = V squared / R (watts)
Series resistance	R total = R1 + R2 + ... ; same current through each; voltages add
Parallel resistance	1 / R total = 1/R1 + 1/R2 + ... ; two resistors: (R1 x R2) / (R1 + R2); same voltage across each; currents add
Kirchhoff	Currents into a node equal currents out; voltages around a loop sum to zero
AC values	**RMS** = 0.707 x peak; peak = 1.414 x **RMS**; peak-to-peak = 2 x peak; **US** mains 120 V **RMS** at 60 **Hz** (many countries 230 V at 50 **Hz**)
Transformer	V secondary / V primary = N secondary / N primary; step-down lowers voltage and raises current; isolation transformer 1:1
Capacitive reactance	Xc = 1 / (2 x pi x f x C); capacitors pass **AC**, block **DC**
Inductive reactance	XL = 2 x pi x f x L; inductors pass **DC**, oppose changes in current
RC time constant	tau = R x C; capacitor reaches 63% of full charge in one tau, about 99% in five
Frequency and period	f = 1 / T; 60 **Hz** has a period of 16.7 **ms**
Stored energy	E = one half x C x V squared (joules); how a defibrillator capacitor stores its charge
Energy	E = P x t; 60 W for 8 h = 0.48 **kWh**; **kWh** is energy, **kW** is power
Voltage divider	V across R1 = V source x R1 / R total (series); 120 V across 60 and 100 ohms puts 45 V on the 60
Current divider	I through a branch = I total x (1/R branch) / (sum of 1/R); the smallest resistance takes the most current

FORMULA	STATEMENT
Kirchhoff's laws	**KCL**: current into a node equals current out (nodal analysis). **KVL**: voltages around a loop sum to zero (mesh analysis)
Internal resistance	(open-circuit voltage minus loaded voltage) / load current; 1.58 V to 1.46 V into 80 ohms = 6.58 ohms; rises with age
Wheatstone bridge	Balanced (no meter current) when R1 / R2 = R3 / R4; strain gauges and thermistors sit in bridges
Complex impedance	Z = R + jX; 40 mH at 60 **Hz** is about 15 ohms of reactance; an ideal 30 mH inductor at 60 **Hz** is 11.3j ohms
AC power	S squared = P squared + Q squared; P true (W), Q reactive (**VAR**), S apparent (**VA**); 180 W and 240 **VAR** give 300 **VA**; power factor = P / S
Op-amp gain	Inverting gain = R feedback / R input (gain 9 with 5 k in needs 45 k feedback); non-inverting = 1 + Rf / Rin
Transformer ratio	V primary / V secondary = N primary / N secondary; 240 V to 6,000 V is a 0.04 ratio
C-rate	Charge or discharge current relative to capacity; 60 **kWh** at 0.5 C = 30 **kW**

- Resistor color code digits: black 0, brown 1, red 2, orange 3, yellow 4, green 5, blue 6, violet 7, gray 8, white 9; tolerance gold 5%, silver 10%
- **US** cord colors: black (or red) hot, white neutral, green or bare ground. **IEC** cords: brown live, blue neutral, green-yellow ground
- Schematic symbols: zigzag resistor, parallel lines capacitor, coil inductor, triangle-with-bar diode, ground, fuse, switch, battery; signals flow left to right; compare measured voltages to the values printed at test points
- Passive devices (resistors, capacitors, inductors, transformers) cannot add energy; active devices (diodes, transistors, **ICs**, op-amps) control or amplify
- Diode conducts one way, forward drop about 0.7 V silicon; zener regulates; **LED** lights; rectifiers: half-wave, full-wave, bridge; capacitor filters ripple; regulator holds output steady
- Linear power supply: heavy transformer, low noise, inefficient. Switching supply: light, efficient, more noise, needs a load
- Transistor: **NPN** or **PNP**, base controls collector-emitter; used as a switch or amplifier. **SCR** and triac switch **AC** power
- Op-amp rules: huge gain, inputs draw no current, feedback makes the inputs equal; instrumentation amplifier rejects common-mode noise (high **CMRR**) for **ECG** front ends; isolation amplifiers and optocouplers keep patient circuits floating
- Filters: low-pass keeps slow signals, high-pass removes baseline drift, band-pass keeps a range, notch removes 60 **Hz**
- **DMM** measures volts, amps (in series), ohms (power off); oscilloscope shows voltage against time, 10x probe divides by ten; function generator provides test signals
- Passive: resistor, capacitor, inductor, memristor, thermistor (no gain). Active: transistor, op-amp, **IC**, **SCR**. Transducer converts energy: sensor to electrical, actuator from electrical; piezoelectric crystals do both
- **US** distribution: step-up to transmit, step-down to use; split-phase 120/240 V at 60 **Hz** with a grounded center neutral; **NEC** is **NFPA** 70 (building wiring), **NESC** is **IEEE** (utility lines)
- Battery specifications: voltage per cell, capacity (Ah), energy density (sets weight), internal resistance (rises with age, sag under load), shelf life, discharge curve, cycle life, C-rate
- Regulators: linear drops excess as heat (quiet, inefficient); switching chops at a duty cycle (efficient, noisy); shorted switching element passes full input voltage, open element gives zero; test regulator input and output pins to ground
- Filters by what they pass: low-pass, high-pass, band-pass, band-stop (notch is a narrow band-stop)

POWER STORAGE AND CONDITIONING	FACTS
Cell voltages	Lead-acid 2.0 V, nickel-cadmium 1.2 V, nickel-metal hydride 1.2 V, lithium-ion 3.6 to 3.7 V, alkaline 1.5 V
Battery behavior	NiCd memory effect (condition with full cycles); Li-ion no memory but thermal runaway risk and must not over-discharge; sealed lead-acid sulfates if left discharged; capacity in amp-hours falls with age
UPS types	Standby (switches on loss), line-interactive (regulates voltage), online double-conversion (always on inverter, best protection)
Power problems	Sag or brownout (low voltage), surge (high), transient spike, noise, blackout; conditioners regulate and filter
Hospital essential electrical system (NFPA 99 and NFPA 70)	Generator restores power within 10 seconds; branches: life safety, critical (patient care, red outlets), equipment; **UPS** bridges the gap for critical devices

POWER STORAGE AND CONDITIONING	FACTS
Isolated power and LIM	Ungrounded secondary of an isolation transformer; line isolation monitor alarms at 5 **mA** total hazard current; used in wet procedure locations
GFCI	Trips at 4 to 6 **mA** of imbalance between hot and neutral in about 25 **ms**; wet locations

04 Electrical safety

CURRENT THROUGH THE BODY (60 HZ, HAND TO HAND)	EFFECT
1 mA	Threshold of perception (tingle)
5 mA	Accepted maximum harmless current
10 to 20 mA	Let-go threshold: muscles contract and the person cannot release
50 mA	Pain, possible fainting, respiratory arrest
100 to 300 mA	Ventricular fibrillation likely
6 A and above	Sustained heart contraction, burns, respiratory paralysis
Microshock	As little as 10 microamps delivered directly to the heart (through a catheter or pacing wire) can cause fibrillation

IEC 60601-1 LIMIT	NORMAL CONDITION	SINGLE FAULT CONDITION
Earth (ground) leakage	5 **mA**	10 **mA**
Touch (enclosure, chassis) current	100 microamps	500 microamps
Patient leakage, type B and BF, AC	100 microamps	500 microamps
Patient leakage, type CF, AC	10 microamps	50 microamps
Patient leakage, DC (all types)	10 microamps	50 microamps
Mains voltage on the applied part	**BF**: 5 **mA**; **CF**: 50 microamps	(single fault test)

NFPA 99 AND FACILITY FACTS	VALUE
Ground (chassis to plug ground pin) resistance	0.5 ohm maximum for patient-care equipment
Chassis (touch) leakage, cord-connected patient-care equipment	500 microamps in current editions (older editions used 300 or 100 microamps; know the limit your facility cites)
Patient care vicinity	About 6 feet (1.8 m) beyond the bed or chair and 7.5 feet (2.3 m) above the floor
Line isolation monitor alarm	5 **mA** total hazard current
GFCI trip	4 to 6 **mA** (nominal 5 **mA**)
Hospital-grade receptacle	Green dot; test retention force, polarity, ground continuity
Relocatable power taps	Only where allowed, rated, and secured; never daisy-chained; no extension cords for patient care equipment
Relocatable power tap load	No more than 75% of the tap's rating (an **AWG** 10 tap rated 35 A carries at most 26 A); never daisy-chained
Prohibited	Three-to-two prong (cheater) adapters; tinned stranded wire ends under screw terminals; cords without strain relief; isolated-ground receptacles in the patient care vicinity

NFPA 99 AND FACILITY FACTS	VALUE
Patient room receptacles	At least 8 per general bed (more in critical care), non-locking, tamper-resistant; no minimum in bathrooms; **OR** receptacles single, duplex, or quadruplex with insulated grounds, locking or not
Household electronics	Allowed only outside the patient care vicinity unless double insulated; inspected before patient use; worn units removed
Gas cylinder storage	Unwrapped, secured upright, minus 7 to 52 C; oxidizers 20 ft from flammables (5 ft with sprinklers, none in a compliant cabinet); no smoking within 20 ft; not in tight enclosures; door warning for other positive-pressure gases
Medical gas systems	**DOT**/**TC**/**ASME** cylinders; gas-specific outlets, no adapters; redundant supply; alarms; labeled zone shutoff valves; inventory and inspection
Telecommunications rooms	No data outlet more than 90 m from its room; at least one per floor; each serves at most 20,000 sq ft; interior rooms in disaster areas
Nurse call	Carries device alarms, staff emergencies, code calls, patient requests; audible and visual annunciation; may forward to pagers
Touch current and clean agents	Touch current: chassis leakage reachable by a person via a path other than ground. Clean agents: nonconductive, residue-free extinguishants (nitrogen, **CO2**, noble gases, halocarbons)

Classes, types, symbols, and testing

- Class I: protective earth (three-prong ground). Class II: double insulation, no ground (square-in-square symbol). Internally powered: battery only
- Applied part type B: body, not for direct cardiac use (stick figure). **BF**: body floating, isolated (figure in a box). **CF**: cardiac floating, safe for direct heart connection (heart in a box)
- Defibrillation-proof applied part: the symbol with paddles; withstands a defibrillator discharge
- Macroshock: current through the skin; needs milliamps. Microshock: current bypassing the skin to the heart; microamps
- Safety analyzer sequence: visual inspection and cord check, ground resistance, chassis leakage (normal, reversed polarity, open ground, open neutral), patient lead leakage, lead-to-lead, mains on applied part
- Leakage rises with open ground and reversed polarity; a rising leakage trend on **PM** predicts insulation failure; test at incoming inspection, after repair, and per **PM** schedule

05 Infection control, hazards, fire, gas, radiation, MRI, and standards

Infection control

- Standard (universal) precautions: treat all blood and body fluids as infectious; hand hygiene before and after every patient contact and before touching equipment
- Transmission-based precautions: contact (gown, gloves), droplet (surgical mask), airborne (N95 respirator, negative-pressure room)
- **PPE** donning: gown, mask, eye protection, gloves. Doffing: gloves, eye protection, gown, mask (clean hands between)
- Cleaning removes soil; disinfection kills most microbes; sterilization kills all including spores
- Spaulding: critical items (enter sterile tissue) sterilize; semi-critical (touch mucous membranes) high-level disinfect; non-critical (intact skin) low-level disinfect
- Steam autoclave: 121 C (250 F) at 15 **psi** for 15 to 30 minutes, or 132 to 135 C flash cycles; biological indicators verify; ethylene oxide and hydrogen peroxide plasma for heat-sensitive items
- Clean and disinfect equipment before servicing it; sharps in puncture-resistant containers; **OSHA** bloodborne pathogens standard covers exposure control; report needlesticks immediately
- Universal precautions (1980s): blood and blood-contaminated fluids only, not sweat, tears, nasal secretions, or urine; bleach cleanup; bagged waste. Standard precautions (**CDC** 1996): all body fluids plus body substance isolation; transmission-based precautions supplement them
- Regulated waste: liquid or free-flowing blood, sealed for a licensed medical waste service; dried blood on dressings sealed in a plastic bag with regular trash; broken glass with tools, never hands

- Lab practice: annual biological risk assessment sets the biosafety level; no mouth pipetting; no food, drink, or contact lenses; retractable needles; decontaminate before disposal; eyewash station; hands-free sinks; clean utility room separate from dirty utility; lab coats are not gowns

Hazard communication and signage

- Safety data sheet (**SDS**): 16 sections including identification, hazards, first aid, fire fighting, handling and storage, exposure controls and **PPE**; must be accessible to staff
- **GHS** pictograms: flame (flammable), flame over circle (oxidizer), skull (acute toxicity), corrosion, exclamation mark (irritant), health hazard (silhouette), gas cylinder, environment
- **NFPA** 704 diamond: blue health, red flammability, yellow reactivity, white special (W with a line means water reactive, **OX** oxidizer); scale 0 to 4
- Laser signs state the class and wavelength; classes 1 (safe) through 4 (fire and eye hazard); wavelength-specific eyewear
- Ionizing radiation: trefoil symbol; time, distance, shielding; **ALARA**; inverse square law; lead aprons about 0.5 mm; dosimeters
- **MRI**: zones I (public) through **IV** (magnet room); the magnet is always on; ferromagnetic objects become projectiles; quench vents helium; **MR** Safe, **MR** Conditional, **MR** Unsafe labels
- Patient precaution signs at the door: contact, droplet, airborne, fall risk, latex allergy, no blood pressure on a limb
- HazCom (**OSHA** 29 **CFR** 1910.1200) requires labels, **SDS** access, training, and records; it adopted the **UN** Globally Harmonized System (**GHS**) for classification, pictograms, signal words, and the 16-section **SDS**
- **SDS** order: identification, hazards, composition, first aid, fire fighting, accidental release, handling and storage, exposure controls and **PPE**, physical and chemical properties, stability and reactivity, toxicology, ecology, disposal, transport, regulatory, other
- Toxicity and limits: **LD50** lethal dose for 50%, **LC50** lethal concentration; **PEL** legal limit (**OSHA**); **TLV** recommended (**ACGIH**); **REL** recommended (**NIOSH**); **STEL** 15-minute ceiling; carcinogen lists from **NTP**, **IARC**, **OSHA**
- **GHS** label: identity, manufacturer contact, pictograms, signal word (Danger or Warning only), hazard and precautionary statements; secondary containers need the same; pyrophoric ignites in air within 5 minutes; oxidizer pictogram exists, reducer does not

Fire and medical gas

- Fire triangle: heat, fuel, oxygen. **RACE**: rescue, alarm, confine, extinguish. **PASS**: pull, aim, squeeze, sweep
- Extinguisher classes: A ordinary combustibles, B flammable liquids, C electrical, D metals, K cooking oils
- Oxygen-enriched atmospheres burn faster; no oil or grease on oxygen fittings; cylinders secured upright, capped, stored below 125 F, full separated from empty
- **US** medical gas cylinder colors: oxygen green, nitrous oxide blue, carbon dioxide gray, nitrogen black, medical air yellow, helium brown; pin index and **DISS** fittings prevent wrong connections
- Full E cylinder of oxygen: about 2,000 to 2,200 **psi**, about 660 liters; wall outlets about 50 **psi**; zone valves shut off gas to an area

STANDARDS, REGULATORS, AND ACCREDITATION	WHAT TO KNOW
CMS	Centers for Medicare and Medicaid Services; sets Conditions of Participation hospitals must meet to be paid; can survey directly
Deemed status	Accrediting organizations survey on **CMS**'s behalf: The Joint Commission, **DNV** Healthcare, **ACHC** (formerly **HFAP**), **CIHQ**
The Joint Commission	Environment of Care and Emergency Management standards cover equipment inventory, maintenance strategies, testing, and documentation
FDA	Regulates devices; medical device reporting: user facilities report a device-related death to **FDA** and the manufacturer within 10 work days, a serious injury to the manufacturer within 10 work days; recalls Class I (serious harm), II (temporary harm), III (unlikely harm)
OSHA	Worker safety: hazard communication, bloodborne pathogens, **PPE**, electrical safety in the workplace
ANSI/AAMI EQ56	Recommended practice for a medical equipment management program
ANSI/AAMI EQ89	Guidance for scheduled maintenance and performance verification

STANDARDS, REGULATORS, AND ACCREDITATION	WHAT TO KNOW
ANSI/AAMI EQ93	Medical equipment management vocabulary and terminology
ANSI/AAMI EQ103	Alternative equipment maintenance (**AEM**) programs: how to justify and document maintenance strategies that differ from the manufacturer's
AEM rules (CMS)	Allowed with documented risk assessment by qualified staff; not for imaging or radiologic equipment, lasers, or where the manufacturer or law requires otherwise; new equipment follows the manufacturer until history supports a change
CMMS	Computerized maintenance management system: inventory and asset tags, work orders, **PM** schedules, equipment history, parts, reports (**PM** completion rate, mean time between failures, cost); every activity documented there
After an incident	Remove the device with all accessories and disposables attached, do not clear logs or settings, tag and sequester, document, report; root cause analysis asks why repeatedly
Joint Commission programs	Accredits and certifies by peer survey; publishes National Patient Safety Goals yearly; links quality and risk management; does not set financial benefits
DNV	Accredits hospitals under **NIAHO**, integrating **CMS** Conditions of Participation with **ISO** 9001; annual surveys
ISO 9000 / 9001	Quality management standards; plan-do-check-act; risk-based thinking; customer focus, leadership, process approach
CMS programs	Medicare, Medicaid, **CHIP**; regulates human laboratory testing; publishes yearly core measure sets (clinical quality measures); fights fraud; does not accredit
CAP	College of American Pathologists: nonprofit accreditor of clinical laboratories under **CMS** authority; inspections every two years plus unannounced
SMDA	Safe Medical Devices Act (1990, amended 1992): user facilities report device-related deaths to **FDA** and manufacturer, serious injuries to the manufacturer; annual summaries

06 Physiological monitoring

ECG FACT	VALUE
Einthoven's triangle	Lead I = **LA** minus **RA**; Lead II = **LL** minus **RA**; Lead III = **LL** minus **LA**; Einthoven's law: Lead II = Lead I + Lead III
Augmented and chest leads	aVR, aVL, aVF from the limb electrodes; V1 to V6 across the chest; 12 leads from 10 electrodes
Electrode colors (US, AHA)	**RA** white, **LA** black, **RL** green (ground), **LL** red, V brown ("white on right, smoke over fire")
Bandwidth	Monitoring mode 0.5 to 40 **Hz**; diagnostic mode 0.05 to 150 **Hz**
Interference	60 **Hz**: bad electrode contact, unshielded leads, nearby equipment; wandering baseline: respiration or motion; spikes: muscle tremor or loose lead
Lead-off	High impedance detected on an electrode; check gel, skin prep, cable
Simulator test	Patient simulator provides rate, amplitude, and arrhythmias to verify the monitor
Rhythms on the strip	Sinus tachycardia fast regular; atrial (supraventricular) tachycardia rapid consistent narrow complexes; atrial fibrillation irregular with quivering waves, no P; ventricular tachycardia rapid WIDE **QRS**; ventricular fibrillation chaotic; asystole flat (check leads)

ECG FACT	VALUE
Lead views and counts	aVF's positive pole is the left leg; 12 leads from 10 electrodes (4 limb, 6 chest); **IEC** colors **RA** red, **LA** yellow, **LL** green, **RL** black differ from **AHA**; thermal print head tested by printing letters or diagonals; paper speeds 10, 12.5, 25, 50 mm/s test the drive

BLOOD PRESSURE FACT	VALUE
Korotkoff sounds	Auscultatory method: cuff above systolic, deflate; first sound (phase 1) is systolic, sounds muffle then disappear (phase 5) at diastolic; automated devices use oscillometry instead
NIBP method	Oscillometric: cuff inflates above systolic, deflates in steps, **MAP** is the point of maximum oscillation, systolic and diastolic derived
Cuff sizing	Bladder width about 40% of arm circumference, length about 80%; too small reads high, too large reads low
Invasive line	Transducer zeroed to atmosphere and leveled at the phlebostatic axis (fourth intercostal space, midaxillary line)
Dicrotic notch	The small dip on the downslope of the arterial waveform marking aortic valve closure; its absence suggests overdamping
Height error	13.6 cm of water equals 10 **mmHg**; every 13.6 cm the transducer is off level changes the reading about 10 **mmHg**
Damping	Fast-flush (square wave) test: overdamped rounds the wave and reads low systolic; underdamped overshoots and reads high
Transducer sensitivity	5 microvolts per volt per **mmHg** (industry standard)
Sphygmomanometer	Cuff, bulb, valve, manometer (mercury, aneroid, electronic), stethoscope; gauge not at zero: zeroing screw; unstable: movement or placement; reads high: cuff below the heart; reads low: cuff above the heart or an air leak (most common); invasive lines read continuously even when hypotensive

OXIMETRY AND CAPNOGRAPHY	VALUE
Pulse oximeter wavelengths	Red about 650 to 660 nm, infrared about 805 to 940 nm; oxyhemoglobin absorbs more infrared; ratio of pulsatile absorbances gives **SpO2**; high perfusion helps, low perfusion hurts
Errors	Carboxyhemoglobin reads falsely high; methemoglobin trends toward 85%; motion, low perfusion, nail polish, ambient light, dyes
Testing	**SpO2** simulator or optical tester; check probe **LED** and detector, cable, and site
Capnography	Infrared absorption by **CO2** (about 4.26 micrometers); mainstream sensor at the airway, sidestream samples through tubing (delay, water trap)
Capnogram	Baseline near zero, rapid rise, alveolar plateau, **EtCO2** at end of plateau, drop on inspiration; loss of waveform means disconnect, apnea, or esophageal intubation

Other monitoring

- Temperature: thermistor (resistance falls as temperature rises, **NTC**), thermocouple (voltage from two metals), infrared tympanic; **YSI** 400 and 700 series probe curves are not interchangeable
- Respiration on a monitor: impedance pneumography through the **ECG** electrodes; capnography is more reliable
- Alarm priorities (**IEC** 60601-1-8): high red flashing, medium yellow flashing, low cyan or yellow steady; alarm fatigue is a patient safety problem; alarm management programs set limits per unit
- Telemetry: **WMTS** bands 608 to 614 **MHz**, 1395 to 1400 **MHz**, 1427 to 1432 **MHz**; dropouts from antenna coverage, low battery, interference; central station shows all beds
- Fetal monitor: ultrasound Doppler for fetal heart rate (110 to 160 bpm), tocodynamometer (**TOCO**) for contractions by pressure on the abdomen; internal scalp electrode and intrauterine pressure catheter are the direct methods
- Respiration methods: chest impedance through **ECG** electrodes (most common), respiratory belts, airflow sampling (thermistor or capnograph), magnetometry; a fingertip oximeter that shuts off mid-use has a low battery
- **EEG** 10-20 landmarks: nasion (bridge of the nose) to inion (back of the skull); beta above 13 **Hz** awake and active; differential amplifier removes common signals

- **EEG** bands: delta under 4 **Hz**, theta 4 to 8, alpha 8 to 13, beta above 13; electrodes placed by the 10-20 system; **EMG** measures muscle electrical activity; evoked potentials time the response to a stimulus

07 Test equipment and units of measure

TEST EQUIPMENT	VERIFIES
Electrical safety analyzer	Ground resistance, chassis and patient lead leakage under normal and fault conditions
Patient (physiological) simulator	**ECG** rate, amplitude, arrhythmias, respiration, invasive pressure, temperature
SpO2 simulator	Saturation and pulse readings through the probe
NIBP simulator	Cuff pressures and oscillometric readings; leak test
Defibrillator analyzer	Delivered energy into 50 ohms, charge time, synchronization, pacing output
ESU analyzer	Output power into a load, leakage, return electrode monitoring
Gas flow and ventilator analyzer	Volume, flow, pressure, oxygen concentration
Infusion pump analyzer	Flow rate accuracy, occlusion pressure
DVM or DMM and meters	Voltage, current, resistance; oxygen analyzer, pressure meter, tachometer, light meter, sound level meter
Cable tracer or toner	Finds the far end of a network cable
What each analyzer does not do	Ventilator analyzer: flow, pressure, volume, O2, breath detection, no **CO2** concentration. **ESU** analyzer: **RMS** power, crest factor, peak-to-peak voltage, **RF** leakage, **REM** thresholds, no handpiece continuity or chassis leakage (multimeter and safety analyzer). Defibrillator analyzer: energy, charge time, sync, not internal impedance. Simulator: no current. **DVM**: voltage only, draws a little current; **DMM** continuity = low resistance, unpowered

UNIT	CONVERSION OR MEANING
mmHg	Blood and gas pressure; 760 **mmHg** = 1 atmosphere; 1 **mmHg** = 1.36 cm **H2O**
cm H2O	Ventilator and airway pressures; 1 cm **H2O** = 0.74 **mmHg**
psi	Gas cylinders and pipelines; 14.7 **psi** = 1 atmosphere; 1 **psi** = 51.7 **mmHg**; 1 **psi** = 6.9 kPa
bar	1 bar = 100 kPa = 14.5 **psi** = 750 **mmHg**, about 1 atmosphere
kPa	1 kPa = 7.5 **mmHg**; 101.3 kPa = 1 atmosphere
Fahrenheit and Celsius	F = C x 1.8 + 32; C = (F minus 32) / 1.8; 37 C = 98.6 F; 0 C = 32 F; 100 C = 212 F
Joules	Energy: 1 J = 1 watt-second; defibrillator output
Lumens and lux	Lumens: total light output of a lamp; lux: lumens per square meter at the surface (surgical lights are rated in lux at 1 m)
Decibels	Sound level and audiometer hearing threshold (**dB HL**); 10 **dB** is a tenfold power ratio
Weight and length	1 kg = 2.2 lb; 1 inch = 2.54 cm; 1 L = 1000 **mL**; 1 **mL** = 1 cc

UNIT	CONVERSION OR MEANING
Prefixes	kilo 10 to the 3, milli 10 to the minus 3, micro 10 to the minus 6, nano 10 to the minus 9

08 Diagnostic, infusion, therapeutic, and laboratory equipment

DIAGNOSTIC DEVICE	FACTS
Otoscope, ophthalmoscope	Battery or wall-powered light and optics for ear and eye; check lamp, battery, charging contacts, specula
Audiometer	Hearing thresholds in **dB HL** at 250 to 8000 **Hz**; calibrate with a sound level meter and coupler; quiet room or booth
Spirometer	**FVC**, **FEV1**, **FEV1**/**FVC** ratio (normal about 0.7 to 0.8); calibrate with a 3 L syringe; filters and mouthpieces
Stress test system	Treadmill or bike with 12-lead **ECG** and **NIBP**; emergency stop; treadmill speed and grade calibration; defibrillator nearby
Scales	Calibrate with certified test weights; zero with the bed or chair empty; load cells
Ultrasound	Piezoelectric transducer sends and receives; 2 to 15 **MHz**; higher frequency, better resolution, less penetration; Doppler measures flow; gel removes air; transducer crystals crack if dropped
Otoscope parts	Handle, head, lamp with collar, magnifying lens, specula, insufflation port; no microphone; loose head cuts light off, worn fibers dim it
Ophthalmoscope	Views the fundus; lens disc, filters, mirror, condensing lens; clean with lens wipes, alcohol discolors
Audiogram	Threshold in **dB** versus frequency; air conduction (headphones) versus bone conduction (mastoid vibrator) separates sensorineural from conductive loss; creates tones, does not measure frequencies
Uroflow meter	Flow rate and volume of voiding (bladder emptying); funnel, beaker, transducer; wireless within about 3 m
Tympanic thermometer	Infrared from the eardrum onto a thermopile; probe covers; 30 minutes to settle; ambient 10 to 40 C; torn cover reads high, cracked lens or obstruction reads low
Endoscope	Fiber-optic or camera, cool light source, insufflation, irrigation, suction; leak test before immersion; no wireless in flexible scopes
Imaging	X-ray: electrons from a tungsten filament strike a tungsten anode; radiopaque bone is white. Fluoroscopy: live X-ray through an image intensifier. **CT**: rotating tube and detectors, computed slices. **MRI**: magnet aligns hydrogen, **RF**, realignment signal, no ionizing radiation, pacemakers contraindicated. **PET**: **FDG** tracer, positron annihilation, two gamma photons. Ultrasound Doppler: toward the probe = higher pitch

INFUSION DEVICE	FACTS
Large-volume pump	Peristaltic; accuracy typically within 5%; rate in **mL**/hr; free-flow protection; occlusion and air-in-line alarms; drug library
Syringe pump	Precise small volumes; syringe size must be recognized; start-up delay and siphoning risks
PCA pump	Patient button, bolus dose, lockout interval, hourly limit; locked to prevent tampering
Feeding (enteral) pump	Enteral-only connectors on purpose; lower accuracy acceptable
Contrast injector	Programmed volume and flow for imaging; pressure limit; air detection; heated syringe
Rate math	**mL**/hr = total **mL** / hours; drops per minute = (**mL**/hr x drop factor) / 60; percent error = (delivered minus set) / set x 100
Enteral versus parenteral	**TPN** by vein through a standard pump; enteral by rotary peristaltic head on wide tubing; low flow = empty bag, occlusion, drip chamber seated wrong or coated; no set = seating, wrong set, or bright light on the detector

INFUSION DEVICE	FACTS
Syringe pump settings	Must know the syringe diameter and brand; small volume, fine control; runs past empty = sensor, size setting, or multi-phase program, not low voltage
PCA settings	Bolus, lockout interval, maximum dose per hour or 4 hours, optional background; logged; locked; ignores requests = check settings first
Infusion modes	Continuous; intermittent (high-rate periods separated by low-rate intervals); bolus; secondary or piggyback; pressure infusers squeeze a bag and are not pumps

THERAPEUTIC DEVICE	FACTS
Infant incubator and warmer	Air and skin temperature servo control, humidity, oxygen; radiant warmer with skin probe; over-temperature alarms; phototherapy blue light about 425 to 475 nm
Patient temperature management	Hypo- and hyperthermia units circulate water through blankets or pads; water temperature limits about 4 to 42 C; check flow, leaks, temperature sensor
Aspiration (suction)	Regulators in **mmHg**, continuous or intermittent; adult range about 80 to 120 **mmHg**; check filter, collection canister, tubing, vacuum source
Sequential compression device	Inflates leg sleeves in sequence to prevent clots; check pressures and cycle timing, hoses, sleeves
Physical therapy equipment	**TENS** and muscle stimulators, therapeutic ultrasound (1 to 3 **MHz**), diathermy, traction, continuous passive motion; output and timer verification
Radiant warmer versus incubator	Radiant: open bed, overhead heater, skin sensor servo, easy access, drafts and dehydration. Incubator: enclosed, air, humidity, oxygen control. Fluctuating temperature = sensor attachment
Ultrasound therapy	1 to 3 **MHz** deep heat for pain and blood flow; needs gel and motion; head too cold is not detected; overheating shutdown = too little gel; slow warming = low drive voltage
Hypo/hyperthermia flow	Heating needs flow of about 1 L/min or more, correct limits, and a good element; poor heating and cooling with no leaks = insufficient flow
Aspirator parts	Vacuum pump, gauge or regulator, tubing, collection container with overflow float, hydrophobic bacterial filter; pulmonary, gastric, surgical; fading suction = leak or filter; sudden stop = fuse or overheated pump
SCD and bilirubin	**SCD** inflates distal (ankle) to proximal (thigh), holds seconds, rests about a minute; high pressure = sleeve too tight. Bilirubin light 360 to 520 nm, blue-green most effective, brighter, closer, more skin; bulb, ballast, fan filter

LABORATORY DEVICE	FACTS
Centrifuge	Speed in **RPM**, force in **RCF** (g); balance opposite loads; lid interlock; verify speed with a tachometer and timer
Incubator	Temperature (often 37 C) and sometimes **CO2** and humidity; verify with a reference thermometer; door seals
Rocker and shaker	Speed and tilt; motor and belt
Refrigerator and freezer	Blood and vaccine storage 2 to 8 C; freezers minus 20 or minus 80 C; continuous temperature logging and alarms; door gaskets and defrost
Microscope	Light source, objectives, focus; cleaning optics
Water bath	Gentle heating under 100 C with stirring; temperature control and uniformity via thermostat and heater; incubates microbial samples; not heating = setpoint or limit below ambient, or low water
Analyzers	Reagents, calibration, quality control samples, fluidics and clogs, temperature
Cryostat and microtome	Cryostat cuts frozen sections in a cold chamber (about minus 20 C); microtome cuts thin sections; blade safety, temperature, advance mechanism

LABORATORY DEVICE	FACTS
Blood gas analyzer	Clark electrode pO2 (platinum cathode, silver anode, O2 membrane, current); glass electrode **pH** (voltage); Severinghaus electrode pCO2 (**pH** electrode behind a **CO2** membrane in bicarbonate); too cold = thermistor; barcode = window
Spectrophotometer	Absorbance at selected wavelengths; tungsten lamp for visible, deuterium for **UV**; read a blank first; quartz cuvettes for **UV**; intermittent at all wavelengths = power cable
Hematocrit centrifuge	Packs red cells in capillary tubes at about 10,000 to 15,000 g; **RCF** rises with speed squared and radius
Microtome detail	Paraffin-embedded blocks; thickness depends on blade clamping, sharpness, clearance angle, cool block, slow steady stroke, no wax; cryostat frost = drafts or breath, splinter = too cold

09 Perioperative and life-support devices

ELECTROSURGERY FACT	VALUE
Frequency	About 300 **kHz** to 3 **MHz**, above the frequency that stimulates nerves and muscle
Modes	Cut: continuous sine wave, vaporizes. Coag: interrupted or damped bursts, dries and seals. Blend mixes them
Monopolar	Current flows from the active electrode through the patient to a large return (dispersive) electrode; return electrode contact quality monitoring prevents burns
Bipolar	Current flows between the two tips of the forceps; no return pad needed
Burns	Poor return pad contact, pad on bony or hairy site, alternate paths through **ECG** electrodes or metal; capacitive coupling in laparoscopy
Video integration	Cameras, light sources, monitors, recording and routing systems; check cable, connector, light cable fibers, white balance
Waveform and faults	Cut = continuous sine, 100% duty cycle, monopolar; coag = pulsed or damped, low duty cycle, higher peak; on but no output = blown high-voltage fuse; will not turn on = cord, fuse, low-voltage supply, switch; footswitch dead = footswitch board then control board

Other perioperative equipment

- Pneumatic tourniquet: cuff pressure above systolic (commonly 250 **mmHg** arm, 300 **mmHg** leg or adjusted to the patient), time limit alarms (about 2 hours), leak test, calibrated gauge
- Fluid and blood warmer: keeps fluids near body temperature; over-temperature cutoff around 41 to 42 C; check heater, sensor, alarms
- Sterilizer: steam 121 C at 15 **psi** or 132 C flash; verify temperature, pressure, time, gasket, chamber; biological indicator; ethylene oxide and hydrogen peroxide for heat-sensitive items
- **OR** table: hydraulics or motors, controls, brakes, weight rating; surgical lights: lux at 1 m, color temperature, handles, bulbs or **LEDs**; surgical microscope: optics, illumination, balance, focus motors
- Sterilization methods: steam (fastest; dry heat for items that cannot be wet), ethylene oxide gas (heat- and moisture-sensitive, about 10 h plus aeration), liquid chemical (small hard-to-reach lumens, rinse), ionizing gamma or beta (disposables in sealed packages); no steam = water pump, low fluid pressure, breaker, heater
- Surgical lasers: **CO2** far infrared absorbed by water for skin and soft tissue; Nd:**YAG** near infrared penetrates (hair, tattoo, coagulation); argon blue-green for pigment and eye; excimer **UV** for corneas; aiming beam and articulated arm; aiming only = footswitch
- Blood warmer: countercurrent warm water around the blood line; pressurized-flow units add a compressor (loud = unseated tubing or failing compressor; hot = blocked vents); prevents transfusion hypothermia, not for lab samples or infants

- Heart-lung machine: venous reservoir, roller or centrifugal pump (gentle on cells), membrane oxygenator (older bubble), heat exchanger for hypothermia, filters, arterial return; heparin thins blood; cardioplegia stops the heart; backup pumps
- Hemodialysis order: access, arterial monitor, blood pump, heparin pump, dialyzer (countercurrent dialysate across a semipermeable membrane), venous monitor, air trap and detector (last), return

DEFIBRILLATION AND PACING	VALUE
Adult energy	Monophasic 360 J; biphasic 120 to 200 J (device specific); pediatric energies lower and weight based
Synchronized cardioversion	Shock timed to the R wave for an organized rhythm the clinician chooses to cardiovert; unsynchronized for **VF** and pulseless **VT**
Testing	Defibrillator analyzer with a 50 ohm load; check delivered energy, charge time, synchronization, pacing output; daily user checks
AED	Analyzes rhythm, advises shock only for **VF** and pulseless **VT**
External pacing	Demand mode paces only when intrinsic rate falls below the set rate; fixed mode paces regardless; output in **mA**; capture confirmed by **QRS** after each spike
Intra-aortic balloon pump	Balloon in the descending aorta inflates in diastole (raises coronary perfusion) and deflates just before systole (lowers afterload); timed to the **ECG** or arterial waveform; helium gas; alarms for timing, leak, and gas loss
Pacing routes	Transcutaneous: chest pads with gel, tens of **mA**, painful, fastest. Transvenous: lead through a vein (subclavian) into the right ventricle, low output. Epicardial: wires on the heart surface at surgery, no subclavian leads. External generators run on **AC** or battery

VENTILATOR AND ANESTHESIA	VALUE
Tidal volume and minute volume	6 to 8 **mL**/kg ideal body weight; minute volume = tidal volume x rate
Typical settings	Rate 12 to 20; I:E ratio 1:2; **PEEP** 5 cm **H2O**; **FiO2** 0.21 to 1.0
Modes	Volume control delivers a set volume; pressure control a set pressure; assist-control, **SIMV**, pressure support, **CPAP**; **BiPAP** has separate inspiratory and expiratory pressures
Alarms	High pressure: obstruction, kink, coughing, secretions. Low pressure or low volume: disconnect or leak. Apnea: no breath detected
Oxygen analyzer calibration	Room air 21% and 100% oxygen; galvanic cells wear out; concentrators deliver about 90 to 95%
Anesthesia machine	Pipeline (50 **psi**) and cylinder supply, flowmeters, agent-specific vaporizers (sevoflurane yellow, isoflurane purple, desflurane blue), hypoxic guard keeps oxygen at least 21 to 25%, oxygen flush, **CO2** absorber, scavenging removes waste gas, pin index and **DISS** prevent misconnection; leak test before use
Anesthesia gas path	Supply (pipeline 50 **psi**, cylinders), regulators, flow controls and flowmeters, vaporizer (first section after the agent's flow control), common gas outlet, breathing circuit with **CO2** absorber, ventilator, patient; scavenging removes waste

10 Problem solving and troubleshooting

The method, in order

- 1. Confirm the complaint: talk to the user, reproduce the problem, rule out use error (wrong settings, wrong accessory, wrong technique), and check the right disposables.
- 2. Check the simple things: power, plug, battery, fuse, cables, connectors, settings, sensors.
- 3. Isolate: which subsystem (power, sensor, signal path, processor, display, mechanical); half-split the path with a simulator and test equipment; localized device or system-wide?

- 4. Form and test one hypothesis at a time; swap known-good parts; use the schematic.
- 5. Repair or replace; follow the service manual.
- 6. Verify: full performance test and electrical safety test after any repair, before return to service.
- 7. Document in the **CMMS**: symptom, cause, action, parts, test results, time. Educate the user if it was a use error.

SYMPTOM	FIRST SUSPECTS
Device dead, no power	Outlet, cord, plug, fuse, power switch, battery, power supply output
Runs on AC but not battery	Battery aged or not charging; charger circuit; battery connector
Intermittent reboot or errors	Failing power supply capacitors, loose connector, overheating
ECG shows 60 Hz noise	Electrode contact, gel, skin prep, lead cable shield, nearby equipment, ground
Wandering baseline	Patient movement or respiration, loose electrode, high-pass filter setting
Intermittent signal	Cable flex failure near connectors, loose connector, cracked solder
NIBP reads high	Cuff too small, cuff too low, patient movement, hose leak causing slow deflation
Arterial line reads low and rounded	Overdamped: air bubble, clot, kinked tubing, loose connection
SpO2 no reading	Probe placement, perfusion, ambient light, probe **LED** failure, cable
Ventilator high-pressure alarm	Kinked circuit, secretions, patient coughing, water in tubing
Ventilator low-pressure or volume alarm	Disconnect, leak, cuff leak, circuit crack
Infusion pump occlusion alarm	Kinked tubing, closed clamp, clotted line, set loaded wrong
Infusion pump air-in-line alarm	Air in set, sensor dirty, set not seated
Defibrillator low delivered energy	Capacitor aging, relay, paddles or pads, battery
ESU patient burn	Return electrode contact, alternate path, damaged cable
Telemetry dropouts on one patient	Transmitter battery, leads, distance; many patients: antenna system or receiver, a system-wide problem
Monitor not sending to the central station or EMR	One bed: cable, port, network config on that monitor. Every bed: switch, server, middleware, network outage
Centrifuge vibration	Unbalanced load, worn rotor or bearing
Refrigerator temperature alarm	Door seal, defrost cycle, compressor, probe, overloading
EtCO2 no value or occlusion	Cannula and tubing, filter present, moisture trap (occlusion test), internal leak (occlude upstream of filter and trap); occlusion: kink, clogged filter, full trap
Respiration rate missing, ECG fine	Input printed circuit board; flat or short trace: raise sensitivity, then electrodes, leads, cable
Blood gas analyzer will not run	Cable and outlet first; 'too cold' after warming = thermistor; no battery run = charger; barcode dead = window
Centrifuge rotor will not spin	Lid not closed firmly (interlock), then connections, board, motor
Microtome sections wrong thickness	Blade clamp, dull edge, clearance angle; compressed = dull, warm block, too fast, wax
Tympanic thermometer reads low	Obstruction or cracked lens (high = torn cover); rest 30 minutes; ambient 10 to 40 C
Enteral pump low flow versus no set	Low flow: empty bag, occlusion, drip chamber. No set: seating, wrong set, bright light
PCA ignores requests	Settings (lockout, maximum) before pendant or motor; 'check settings' = door locked before setup
Aspirator suction fades	Leak (tubing, container, vacuum line) or clogging filter; sudden stop = fuse or overheated pump; none = disconnected, blocked filter, full container
ESU on but no output	High-voltage supply fuse

SYMPTOM	FIRST SUSPECTS
Tourniquet pump runs constantly	Leak in tubing, cuff, or valves, or a bad pressure transducer; will not inflate = pump disconnected
Autoclave no steam	Water pump, low fluid pressure, breaker or overload, burned-out heater, low water; over-pressure = pressure switch
Bilirubin light dark or dim	Too hot to relight; bulb then ballast; dirty lens; overheating = fan filter, fans, thermistor
Device went to battery while plugged in	Outlet and power cable continuity first, then the internal power circuit; regulator: measure input and output pins to ground

Priorities, communication, maintenance

- Priority order: life support and high-risk devices first, then patient risk, then urgency and downtime and clinical impact; loaners for critical downtime
- Device error versus use error: a device that performs to specification but was used wrongly needs education and possibly a design or labeling report, not a repair
- In-service and cross-training: teach proper use, care, cleaning, and alarm response; document attendance; communicate clearly with clinicians, staff, and manufacturers in writing and verbally
- Preventive maintenance: scheduled or risk-based; **AEM** programs allowed with documented justification (not imaging, lasers, or where the manufacturer requires); **PM** after repair; performance verification against tolerances
- Calibration traceable to national standards; test equipment has its own calibration schedule
- Incoming inspection before first use: physical, functional, safety test, inventory entry, manuals on file; lifecycle ends with data wiped, decontamination, and disposition
- Criticality = importance to the mission + likelihood of failure or **PM** demand + potential harm; repair time is not a factor; redundancy lowers priority but does not prevent all downtime; checklists guide user inspections
- Root cause analysis: what happened, why, how to prevent; keep asking why (five whys); ultrasound blurry = crystal then gel; oximeter erratic = ambient light; one sphygmomanometer = technique then cracked tubing (a leak reads low)
- Local versus network: one room missing = that room; several = shared configuration or network; false alarms on many monitors = alarm limits; check physical links first; search repair records for fleet patterns; ping, cable tester, multimeter help, a safety analyzer does not
- Switching supply shorted element = full input voltage out, open = zero; **UPS** short runtime = battery, no backup = inverter, switched over while plugged in = outlet or cable; desktop **PSUs** die of fan failure and heat

11 Healthcare information technology

NETWORKING FACT	VALUE
IPv4 address	Four numbers 0 to 255; subnet mask separates network and host; default gateway is the exit router; same subnet required to talk without a router
MAC address	48-bit hardware address in 12 hex digits, burned into the network card; switches forward by **MAC**
Private ranges	10.0.0.0/8; 172.16.0.0 to 172.31.255.255; 192.168.0.0/16
DHCP and static	**DHCP** assigns addresses automatically; medical devices often use static addresses or **DHCP** reservations so they are always reachable
DNS	Names to addresses; ping by address works but by name fails means **DNS**
APIPA	169.254.x.x means **DHCP** failed
Ports	**HTTP** 80, **HTTPS** 443, **SSH** 22, **DICOM** commonly 104 or 11112, **HL7** commonly 2575 (site specific)
Devices	Switch connects devices in a **LAN** by **MAC**; router connects networks by **IP**; access point connects **Wi-Fi** clients; access point controller manages many **APs**; **KVM** switch shares one keyboard, video, and mouse among servers; virtual servers run many operating systems on one host; **LAN** is local, **WAN** spans sites, cloud is provider-hosted
Cabling	Cat 5e 1 **Gbps**, Cat 6A 10 **Gbps**, 100 m limit; fiber for longer runs; link light off means Layer 1; cable tracer finds the far end

NETWORKING FACT	VALUE
Wi-Fi	2.4 **GHz** longer range, channels 1, 6, 11; 5 **GHz** more channels, less interference; **WPA2** or **WPA3**; enterprise mode uses **802.1X** with **RADIUS**
Telemetry and RTLS	**WMTS** bands for patient telemetry; real-time location systems track equipment with **RF**, infrared, or ultrasound tags
OSI layers	1 physical (media, hubs), 2 data link (frames, **MAC**, switches), 3 network (**IP**, routers), 4 transport (**TCP**/**UDP**, ports), 5 session, 6 presentation (format conversion, encryption), 7 application (**APIs**); no conversion layer
Topologies	Bus and ring fail on any break; star (central switch) survives one bad cable; hybrid; mesh partial or full; point-to-multipoint (central node also processes); physical versus logical
Binary masks	255.255.255.240 = ...11110000 (/28, 16 addresses); 248 = 11111000; 252 = 11111100; ones must be contiguous; subnetting separates departments and limits broadcasts
Hub versus switch, PPP	Hub repeats to every port at layer 1, half duplex, collisions; switch keeps a source address table and forwards at layer 2, full duplex; **PPP** links two routers over serial or dial with authentication

COMMAND OR TOOL	USE
ping	Is the address reachable, and how fast; ping the gateway first
ipconfig (Windows), ifconfig or ip (Linux)	Show the device's address, mask, gateway; release and renew **DHCP**
tracert or traceroute	Every hop to a destination; where the path breaks
nslookup	Test name resolution
arp -a	**IP** to **MAC** table; duplicates
netstat	Open connections and listening ports
Cable tracer and tester	Find the far end; check pinout and continuity
TDR	Time-domain reflectometer: pulse reflection shows whether a cable is open and how far to the break
ping (ICMP)	Echo request and reply, **IPv4** and **IPv6**; reports round-trip **ms**, packet loss, min, max, average, deviation; not a cable resistance test; power cycling is not a test
Fiber inspection microscope	Checks connector ends for dirt and damage; clean with isopropyl alcohol and lint-free wipes or cut and repolish; light source and power meter measure loss

INTEROPERABILITY	MEANING
HL7 version 2	Text messages between systems: **ADT** (admit, discharge, transfer), **ORM** (orders), **ORU** (results, including device data)
HL7 FHIR	Modern web-based standard using resources and **APIs**
DICOM	Standard for medical images and their transfer; modality worklist pulls patient data to the device; **PACS** stores and serves images
EMR and EHR	Electronic medical record (within one organization) and electronic health record (across); devices feed vitals in directly or through middleware
Middleware	Translates and routes between devices and the **EMR**; device gateways; where most integration faults are diagnosed
Time synchronization	**NTP** keeps device clocks aligned so records and alarms line up
PACS, RIS, LIS, LIMS	**PACS** stores and transports images with patient data (redundant servers); **RIS** schedules, tracks, and reports radiology; **LIS** tracks specimens and analyzer results; **LIMS** is the research and high-volume version

INTEROPERABILITY	MEANING
DICOM and HL7 detail	**DICOM** is a 22-part standard for image format and transfer referencing **IEC** definitions; **HL7** standards: version 2 messaging, version 3 **XML**, Clinical Document Architecture; **HL7** does not set leakage limits or breach rules

Protected data and cybersecurity

- **HIPAA**: privacy rule (who may see **PHI**), security rule (administrative, physical, technical safeguards), breach notification; **PHI** is any identifiable health information; minimum necessary
- **NIST** Cybersecurity Framework functions: govern, identify, protect, detect, respond, recover
- **HITRUST**: a certifiable framework combining many requirements; **HHS** cybersecurity performance goals: essential and enhanced practices for hospitals (**MFA**, patching, backups, segmentation, training)
- **MDS2**: manufacturer disclosure statement for medical device security; **SBOM**: software bill of materials
- Controls: unique accounts and **MFA**, least privilege, encryption at rest and in transit, patching on the vendor's validated schedule, network segmentation for devices that cannot be patched, antivirus where the vendor allows, physical security of workstations, screen locks
- Backups: 3 copies, 2 media, 1 offsite; test restores; ransomware response: isolate, report, restore
- Incident: disconnect the device from the network, do not power off if forensics are needed, report to **IT** security and the vendor, document
- **HITECH** (2009): **EHR** incentives and penalties, meaningful use, breach notification to individuals and **HHS**, business associates bound, no sale of **PHI**, patient access; sets no spending guidelines
- **HIPAA** titles: I insurance portability; II administrative simplification with the privacy and security rules, fraud control, and fines; III medical savings accounts; **IV** enforcement by the Office for Civil Rights; V revenue offsets
- **DMCA** (1998): bans circumvention tools, raises penalties, limits provider liability; exemptions for security testing and personal implant data; does not ban copying devices as such
- **FDA MDDS**: systems that store, transfer, reformat, or display device data without controlling devices; reclassified Class III to Class I in 2011; a vital signs monitor yes, a lobby **TV** no
- **IEC** 80001-1 (2010, **IEC** and **ISO**): risk management for **IT** networks with medical devices; excludes drugs; applies to any organization's network; defines no acceptable risk level; not from the **FDA**
- **MDS2** contents: automatic logoff, audit controls, authorization, configurable security, security updates, de-identification, backup and recovery, emergency (break-glass) access, data integrity, malware protection, node and user authentication, physical locks, third-party component plans, remote servicing; not structural integrity

Computer hardware and software troubleshooting

- Hardware: power supply (no power, random reboots), hard drive (slow, errors, clicking; **SMART** warnings; replace and restore image), memory (crashes), peripherals (drivers, cables, ports), connectors and cables (reseat, swap)
- Software: Windows, Office, Linux, and Unix updates and patches only on the vendor's validated schedule for medical devices; drivers after updates; event logs; restore from image; user accounts and permissions
- Order of checks: power and cables, link lights, address and gateway, ping the gateway, ping the server, name resolution, application or service, then escalate to **IT** or the vendor

12 Acronyms

Every acronym used in this course, 288 in all. In lessons and on this sheet, hover over an acronym for the short version; click it, or click one below, for the longer explanation.

ACRONYM	STANDS FOR	MEANING
802.11	IEEE 802.11	The family of Wi-Fi standards.
802.1X	IEEE 802.1X	Port-based network access control: a device must authenticate before the switch port or Wi-Fi passes traffic.
AA	AA cell	A common single-cell battery size, about 1.5 V.
AAMI	Association for the Advancement of Medical Instrumentation	The standards body for medical equipment and the sponsor of the CBET credential.
ABC	Class A, B, and C fire extinguisher	A multipurpose extinguisher for ordinary combustibles, flammable liquids, and electrical fires.
AC	Alternating current	Mains power that reverses direction, 60 Hz in the US and 50 Hz elsewhere.

ACRONYM	STANDS FOR	MEANING
ACGIH	American Conference of Governmental Industrial Hygienists	The organization that publishes threshold limit values.
ACHC	Accreditation Commission for Health Care	An accrediting body that absorbed the HFAP hospital program.
ACI	AAMI Credentials Institute	The body within AAMI that administers the CBET and related certifications.
ACK	Acknowledgment (HL7 message)	The reply an HL7 interface sends to confirm it received a message, carrying an accept or error code.
ADC	Analog-to-digital converter	Turns a continuous voltage into digital samples.
ADT	Admit, discharge, transfer	The HL7 messages that tell every system who is in which bed.
AE	Application entity	A DICOM node, identified by its AE title, IP address, and port.
AED	Automated external defibrillator	A defibrillator that analyzes the rhythm and advises a shock only for VF or pulseless VT.
AEM	Alternative equipment maintenance	A CMS-allowed program that adjusts maintenance from manufacturer recommendations based on risk and history.
AF	Atrial fibrillation	An irregular rhythm with no P waves.
AHA	American Heart Association	The organization behind CPR guidelines and the US ECG electrode color code.
ALARA	As low as reasonably achievable	The radiation safety principle: minimize dose with time, distance, and shielding.
ANSI	American National Standards Institute	The US body that accredits standards developers such as AAMI.
AP	Access point	The Wi-Fi radio that bridges wireless clients onto the wired network.
API	Application programming interface	The defined way one program talks to another, such as the web endpoints a cloud service exposes.
APIPA	Automatic Private IP Addressing	The 169.254.x.x address a host gives itself when it cannot reach a DHCP server.
ASME	American Society of Mechanical Engineers	The standards body for pressure vessels and boilers.
AV	Atrioventricular node	The relay between atria and ventricles that delays the impulse so the ventricles fill.
AWG	American Wire Gauge	The wire size scale; smaller numbers are thicker wire with higher ampacity.
BAA	Business associate agreement	The HIPAA contract a vendor signs before handling protected health information.
BF	Body floating (applied part)	An applied part that touches the patient but not the heart, isolated from ground; stick-figure-in-a-box symbol.
BIOS	Basic Input/Output System	The original PC firmware: 16-bit, text setup, MBR disks up to 2 TB.
BiPAP	Bilevel positive airway pressure	Two pressure levels, higher on inhalation and lower on exhalation, for non-invasive ventilation.
BP	Blood pressure	The pressure in the arteries, systolic over diastolic, in mmHg.
C-ECHO	DICOM echo service	The DICOM ping that verifies two application entities can communicate.
CAP	College of American Pathologists	The organization that accredits clinical laboratories.
CBET	Certified Biomedical Equipment Technician	The AAMI credential for biomedical equipment technicians.
CDA	Clinical Document Architecture	HL7's XML standard for clinical documents such as discharge summaries.
CDC	Centers for Disease Control and Prevention	The US public health agency that sets infection control guidance such as standard precautions.
CF	Cardiac floating (applied part)	An applied part that may connect directly to the heart, with the strictest leakage limits; heart-in-a-box symbol.
CFR	Code of Federal Regulations	The codified US federal rules; 29 CFR 1910 holds OSHA standards, 21 CFR the FDA's.
CHIP	Children's Health Insurance Program	Federal and state health coverage for children, run by CMS.
CIA	Confidentiality, integrity, and availability	The three goals of security: keep data secret, keep it correct, keep it reachable.
CIHQ	Center for Improvement in Healthcare Quality	A newer CMS-approved hospital accreditor.
CMMS	Computerized maintenance management system	The database that tracks equipment inventory, work orders, and preventive maintenance.
CMOS	Complementary metal-oxide semiconductor	The small battery-backed memory that keeps firmware settings and the clock.
CMRR	Common-mode rejection ratio	How well an amplifier cancels signals present on both inputs, such as 60 Hz interference.
CMS	Centers for Medicare and Medicaid Services	The federal agency whose Conditions of Participation hospitals must meet to be paid.
CO	Carbon monoxide	A toxic gas that binds hemoglobin and makes pulse oximeters read falsely high.
CO2	Carbon dioxide	The gas exhaled as the waste product of metabolism; measured by capnography, and the lasing medium of CO2 surgical lasers.
COPD	Chronic obstructive pulmonary disease	Long-term airway obstruction, usually from smoking, with a low FEV1/FVC ratio.
CPAP	Continuous positive airway pressure	A constant pressure applied to the airway of a spontaneously breathing patient.
CPR	Cardiopulmonary resuscitation	Chest compressions and breaths to keep blood moving during cardiac arrest.
CPU	Central processing unit	The processor that executes instructions.
CT	Computed tomography	Cross-sectional X-ray imaging from a rotating tube and detector array.
DAC	Digital-to-analog converter	Turns digital values back into a voltage to drive displays, speakers, or outputs.
dB	Decibel	A logarithmic ratio unit for signal levels, gain, and sound.
DBP	Diastolic blood pressure	The arterial pressure between beats, while the heart fills.

ACRONYM	STANDS FOR	MEANING
DC	Direct current	Current in one direction: batteries and power supply outputs.
DHCP	Dynamic Host Configuration Protocol	Hands out IP addresses, masks, gateways, and DNS servers automatically (UDP 67 and 68).
DICOM	Digital Imaging and Communications in Medicine	The standard for medical image files and their network exchange, by convention on port 104.
DISS	Diameter Index Safety System	Gas-specific threaded fittings that make it impossible to connect the wrong gas hose.
dL	Deciliter	One tenth of a liter; lab values such as glucose in mg per dL.
DMCA	Digital Millennium Copyright Act	The 1998 copyright law that bans circumventing digital protections.
DMM	Digital multimeter	The handheld meter for voltage, current, resistance, and continuity.
DNA	Deoxyribonucleic acid	The genetic material; sterilization kills microbes by destroying it.
DNS	Domain Name System	Translates names such as www.example.com into IP addresses (UDP 53, TCP 53 for zone transfers).
DNV	DNV Healthcare	An accrediting organization that surveys hospitals annually under its NIAHO program.
DOT	Department of Transportation	The US agency whose rules govern compressed gas cylinders.
DVM	Digital voltmeter	A meter for voltage; in practice the same instrument as a DMM.
ECG	Electrocardiogram	The recording of the heart's electrical activity: P wave, QRS complex, T wave.
ECRI	ECRI Institute	The independent organization that tests devices and publishes hazard alerts and the annual top-ten hazards list.
EDI	Electronic data interchange	Standardized electronic exchange of business documents such as claims and payments.
EEG	Electroencephalogram	The recording of the brain's electrical activity from scalp electrodes, in microvolts.
EHR	Electronic health record	The patient's longitudinal digital record, shared across providers.
EM	Electromagnetic	Electric and magnetic fields; EM interference disrupts monitors and telemetry.
EMG	Electromyogram	The recording of muscle electrical activity.
EMI	Electromagnetic interference	Noise from motors, power lines, fluorescent lights, and radios that corrupts signals on copper cable.
EMR	Electronic medical record	The hospital's clinical record system: orders, notes, results, medications.
EQ	AAMI EQ series	AAMI's equipment management standards: EQ56, EQ89, EQ93, EQ103.
EQ103	ANSI/AAMI EQ103	Guidance for alternative equipment maintenance programs.
EQ56	ANSI/AAMI EQ56	The recommended practice for a medical equipment management program.
EQ89	ANSI/AAMI EQ89	Guidance for scheduled maintenance and performance verification.
EQ93	ANSI/AAMI EQ93	The vocabulary of medical equipment management.
ESD	Electrostatic discharge	Static electricity that can destroy chips; prevented with a wrist strap, mat, and antistatic bags.
ESU	Electrosurgical unit	The generator that cuts and coagulates tissue with high-frequency current above about 100 kHz.
EtCO2	End-tidal carbon dioxide	The CO2 concentration at the end of exhalation, normally 35 to 45 mmHg, measured by capnography.
ETO	Ethylene oxide	A toxic gas used to sterilize heat-sensitive devices.
FDA	Food and Drug Administration	The US regulator of medical devices; classes I to III, recalls, and medical device reporting.
FDG	Fluorodeoxyglucose	The glucose-like radioactive tracer used in PET scans.
FEV1	Forced expiratory volume in one second	The volume exhaled in the first second of a forced breath out.
FHIR	Fast Healthcare Interoperability Resources	HL7's web-style standard using REST APIs and JSON resources.
FHR	Fetal heart rate	The baby's heart rate, measured by ultrasound Doppler on a fetal monitor.
FiO2	Fraction of inspired oxygen	The oxygen concentration being delivered, from 0.21 (room air) to 1.0.
FSH	Follicle-stimulating hormone	The pituitary hormone that drives sperm production and egg development.
FVC	Forced vital capacity	The total volume a patient can exhale forcefully after a full breath in.
Gbps	Gigabits per second	Billions of bits per second; gigabit Ethernet is 1 Gbps.
GFCI	Ground-fault circuit interrupter	A receptacle or breaker that trips when current leaking to ground exceeds about 5 mA.
GHS	Globally Harmonized System	The international standard for chemical hazard labels and safety data sheets.
GHz	Gigahertz	Billions of cycles per second; CPU clock speed and Wi-Fi band frequency.
GI	Gastrointestinal	The digestive tract.
H2O	Water	The pressure unit centimeters of water (cmH2O) for airway and ventilator pressures.
HDD	Hard disk drive	Spinning magnetic storage: cheap per terabyte, slower, and fragile.
HFAP	Healthcare Facilities Accreditation Program	The former osteopathic hospital accreditation program, now part of ACHC.
HHS	Department of Health and Human Services	The US department that enforces HIPAA through its Office for Civil Rights.
HIPAA	Health Insurance Portability and Accountability Act	The US law protecting health information; Title II holds the privacy and security rules.
HITECH	Health Information Technology for Economic and Clinical Health Act	The 2009 law that funded EHR adoption and strengthened HIPAA enforcement and breach notification.

ACRONYM	STANDS FOR	MEANING
HITRUST	Health Information Trust Alliance	A certifiable security framework that combines HIPAA, NIST, and other requirements for healthcare.
HIV	Human immunodeficiency virus	A bloodborne virus; one of the pathogens the OSHA bloodborne pathogen standard addresses.
HL	Hearing level	The decibel scale audiometers use, referenced to normal hearing thresholds.
HL7	Health Level Seven	The messaging standard that connects clinical systems: version 2 pipes, version 3 XML, FHIR web APIs.
HR	Heart rate	Beats per minute, derived from the ECG or pulse.
HTTP	Hypertext Transfer Protocol	Web traffic in the clear on TCP 80.
HTTPS	Hypertext Transfer Protocol Secure	HTTP inside TLS on TCP 443.
HV	High voltage	The kilovolt sections of X-ray generators, defibrillators, and ESUs.
HVAC	Heating, ventilation, and air conditioning	The building's climate systems, which keep server rooms cool and are themselves networked controls.
Hz	Hertz	Cycles per second; display refresh rate, AC mains frequency, and signal frequency.
IaaS	Infrastructure as a service	The provider supplies virtual machines, storage, and networks; you manage the operating system and up.
IABP	Intra-aortic balloon pump	A balloon in the aorta that inflates in diastole and deflates in systole to support the heart.
IARC	International Agency for Research on Cancer	The WHO body that classifies carcinogens.
IBP	Invasive blood pressure	Pressure measured directly through an arterial line and a transducer.
IC	Integrated circuit	A chip: many transistors on one piece of silicon.
ICMP	Internet Control Message Protocol	The protocol behind ping and traceroute.
ICU	Intensive care unit	The critical care area with the highest density of monitors, ventilators, and pumps.
ID	Identification	A patient identifier or a device identifier.
IEC	International Electrotechnical Commission	The international standards body; IEC 60601-1 is the medical electrical equipment safety standard.
IEEE	Institute of Electrical and Electronics Engineers	The engineering body behind Ethernet (802.3), Wi-Fi (802.11), and many device standards.
IP	Internet Protocol	The Layer 3 protocol that addresses and routes packets between networks.
IPv4	Internet Protocol version 4	32-bit addresses written as four decimal octets, such as 192.168.1.10.
IPv6	Internet Protocol version 6	128-bit addresses in eight hex groups; no broadcast, built-in autoconfiguration.
ISO	International Organization for Standardization; also an ISO disc image	The standards body, and the .iso file format that holds a complete disc image.
IT	Information technology	Computers, networks, and the people who run them.
IUPC	Intrauterine pressure catheter	A catheter that measures contraction pressure directly.
IV	Intravenous	Into a vein; IV lines, pumps, and fluids.
KCL	Kirchhoff's current law	Current into a node equals current out.
kHz	Kilohertz	One thousand cycles per second.
kV	Kilovolt	One thousand volts; X-ray tube voltage and defibrillator charge levels.
KVL	Kirchhoff's voltage law	The voltages around any closed loop sum to zero.
KVM	Keyboard, video, mouse (switch)	One keyboard, monitor, and mouse shared among several computers.
KVO	Keep vein open	A very low infusion rate that keeps an IV line from clotting.
kW	Kilowatt	One thousand watts of power.
kWh	Kilowatt-hour	The energy unit on the electric bill: one kilowatt for one hour.
LA	Left arm electrode (also left atrium)	The black ECG electrode on the left arm; in heart anatomy, the left atrium.
LAN	Local area network	One site's network: a building, floor, or campus that one organization owns and runs.
LC50	Lethal concentration 50 percent	The airborne concentration lethal to half of test subjects.
LCD	Liquid crystal display	The flat panel that shapes light from a backlight into an image; TN, IPS, and VA are the panel types.
LD50	Lethal dose 50 percent	The dose that kills half of test subjects.
LED	Light-emitting diode	A small light source: the status lights on a port, the lamp in cheap fiber transmitters, and modern display backlights.
LH	Luteinizing hormone	The pituitary hormone that drives testosterone and triggers ovulation.
LIM	Line isolation monitor	Continuously measures how much current a fault could drive from an isolated power system to ground and alarms at 5 mA.
LIMS	Laboratory information management system	Sample and workflow tracking software for laboratories.
LIS	Laboratory information system	The software that manages lab orders, analyzer results, and reporting.
LL	Left leg electrode	The red ECG electrode on the left leg.
LV	Left ventricle	The chamber that pumps blood to the body at systemic pressure.

ACRONYM	STANDS FOR	MEANING
LVP	Large-volume pump	An infusion pump for bags of fluid, usually peristaltic, accurate to about 5 percent.
mA	Milliampere	One thousandth of an ampere; the unit for shock thresholds and leakage currents.
MAC	Media access control (address)	The 48-bit hardware address burned into a network interface, used at Layer 2.
MAP	Mean arterial pressure	The average arterial pressure: diastolic plus one third of the pulse pressure.
MDDS	Medical device data system	Software or hardware that transfers, stores, or displays device data without changing it.
MDS2	Manufacturer Disclosure Statement for Medical Device Security	The standard form describing a device's security capabilities.
MFA	Multi-factor authentication	Two or more different kinds of proof of identity.
MHz	Megahertz	Millions of cycles per second; memory speed, radio channels, and older CPU clocks.
mL	Milliliter	One thousandth of a liter; tidal volumes, drug doses, and infusion rates in mL per hour.
mmHg	Millimeters of mercury	The pressure unit for blood pressure, gas pressures, and vacuum.
MR	Magnetic resonance	The MRI environment; equipment is labeled MR Safe, MR Conditional, or MR Unsafe.
MRI	Magnetic resonance imaging	Imaging with a strong magnet and radio pulses; no ionizing radiation, but the magnet is always on.
MRN	Medical record number	The hospital's unique identifier for a patient.
ms	Millisecond	One thousandth of a second; latency, pulse widths, and ECG timing.
MSH	Message header segment	The first segment of every HL7 version 2 message.
mV	Millivolt	One thousandth of a volt; ECG signals are about 1 mV.
NAT	Network address translation	Rewrites private inside addresses to a public address at the router so many hosts can share it.
NEC	National Electrical Code	The US wiring code, NFPA 70.
NESC	National Electrical Safety Code	The utility-side safety code for power lines and substations.
NFPA	National Fire Protection Association	The standards body for fire and electrical safety: NFPA 70 (NEC), NFPA 99 (health care facilities), NFPA 101 (life safety).
NIAHO	National Integrated Accreditation for Healthcare Organizations	DNV's hospital accreditation program.
NIBP	Non-invasive blood pressure	Automatic cuff measurement by the oscillometric method.
NIOSH	National Institute for Occupational Safety and Health	The CDC's research arm for workplace safety, which publishes RELs.
NIST	National Institute of Standards and Technology	The US standards body whose cybersecurity framework hospitals use.
NPN	Negative-positive-negative transistor	A bipolar transistor that conducts when its base is pulled positive.
NPSG	National Patient Safety Goals	The Joint Commission's annual list of safety priorities, including alarm management.
NTC	Negative temperature coefficient (thermistor)	A thermistor whose resistance falls as temperature rises; the sensor in patient temperature probes.
NTP	National Toxicology Program	The US program that publishes the Report on Carcinogens, one of the carcinogen listings cited on a safety data sheet.
NTP	Network Time Protocol	The protocol (UDP 123) that keeps device clocks synchronized so logs, alarms, and records line up.
OBR	Observation request segment	The HL7 segment describing the order a result answers.
OBX	Observation segment	The HL7 segment holding one observation and its value.
OCR	Office for Civil Rights	The HHS office that enforces HIPAA privacy and security rules.
OR	Operating room	The surgical suite, with isolated power, ESUs, anesthesia machines, and lasers.
ORM	Order message	The HL7 version 2 message that carries an order.
ORU	Observation result	The HL7 message that carries results, including vital signs from monitors.
OSHA	Occupational Safety and Health Administration	The US agency that sets workplace safety rules.
OSI	Open Systems Interconnection model	The seven-layer model for describing where a network function happens.
OX	Oxidizer (NFPA 704 symbol)	The special hazard code in the white diamond for oxidizing materials.
PaaS	Platform as a service	The provider runs the platform (OS, runtime, database); you deploy code.
PACS	Picture archiving and communication system	The server that stores and distributes medical images using DICOM.
PASS	Pull, aim, squeeze, sweep	How to use a fire extinguisher.
PC	Personal computer	A desktop or laptop for one user.
PCA	Patient-controlled analgesia	An infusion pump that lets the patient self-dose pain medication within programmed limits.
PCB	Printed circuit board	The board that holds and connects a device's components.
PDCA	Plan, do, check, act	The continuous improvement cycle.
PEEP	Positive end-expiratory pressure	Pressure the ventilator keeps in the lungs at the end of exhalation so alveoli stay open.
PEL	Permissible exposure limit	OSHA's legal maximum exposure to a substance, averaged over a shift.
PET	Positron emission tomography	Nuclear imaging of metabolic activity using a positron-emitting tracer such as FDG.

ACRONYM	STANDS FOR	MEANING
pH	Potential of hydrogen	The acidity scale; blood is normally 7.35 to 7.45.
PHI	Protected health information	Any health information tied to an identifiable person.
PID	Patient identification segment	The HL7 segment carrying patient name, ID, and demographics.
PM	Preventive maintenance	Scheduled inspection and testing of equipment before it fails.
PNP	Positive-negative-positive transistor	A bipolar transistor that conducts when its base is pulled negative.
PoE	Power over Ethernet	Delivers DC power to devices over the data cable: 802.3af 15.4 W, 802.3at 30 W, 802.3bt 60 to 90 W.
PP	Pulse pressure	Systolic minus diastolic pressure.
PPE	Personal protective equipment	Gloves, gowns, masks, eye protection, and lead aprons.
PPP	Point-to-Point Protocol	The Layer 2 protocol for serial and dial-up links that carries IP with authentication.
PR	PR interval	The ECG interval from the start of the P wave to the start of the QRS, 0.12 to 0.20 seconds; the AV node delay.
psi	Pounds per square inch	The pressure unit for cylinders, sterilizers, and pneumatic systems.
PSU	Power supply unit	Converts wall AC to the DC voltages a computer needs: 12 V, 5 V, 3.3 V.
PSV	Pressure support ventilation	The ventilator boosts each spontaneous breath with a set pressure.
PT	Physical therapy	The rehabilitation department, whose equipment includes TENS units, stimulators, and therapeutic ultrasound.
QC	Quality control	Running known samples to verify an analyzer is reading correctly.
QRS	QRS complex	The sharp ECG spike of ventricular depolarization; normally under 0.12 seconds wide.
QT	QT interval	The ECG interval from the start of the QRS to the end of the T wave, under about 0.44 seconds.
RA	Right arm electrode (also right atrium)	The white ECG electrode on the right arm; in heart anatomy, the right atrium.
RACE	Rescue, alarm, confine, extinguish or evacuate	The order of actions when a fire is discovered.
RADIUS	Remote Authentication Dial-In User Service	The AAA server behind Wi-Fi Enterprise, VPNs, and 802.1X; UDP 1812 and 1813.
RAID	Redundant array of independent disks	Combining disks for speed, capacity, or fault tolerance: 0 stripe, 1 mirror, 5 stripe with parity, 6 double parity, 10 mirrored stripes.
RAM	Random-access memory	The working memory that holds running programs and data; lost at power off.
RC	Resistor-capacitor (circuit)	A resistor and capacitor together, whose time constant is R times C.
RCA	Root cause analysis	A structured investigation of what happened, why, and how to prevent it.
RCF	Relative centrifugal force	The g-force a centrifuge applies, which depends on speed and rotor radius.
REL	Recommended exposure limit	NIOSH's recommended exposure limit.
REM	Roentgen equivalent man	A unit of radiation dose equivalent, 100 rem equals 1 sievert.
RF	Radio frequency	Radio signals; the medium of Wi-Fi, cellular, and Bluetooth, and a source of interference and, in medicine, surgical energy.
RIS	Radiology information system	Scheduling, tracking, and reporting for imaging departments.
RJ45	Registered jack 45	The eight-pin plug on Ethernet twisted-pair cables (properly an 8P8C connector).
RL	Right leg electrode	The green ECG electrode on the right leg, the reference or driven ground.
RMS	Root mean square	The effective value of an AC voltage or current, the DC equivalent for heating; 120 V mains is RMS.
RPM	Revolutions per minute	Rotation speed: hard drive platters at 5400 or 7200, fans, and laboratory centrifuges.
RS-232	Recommended Standard 232 serial	The classic serial port: slow, simple, and still used for console access and legacy devices.
RTLS	Real-time location system	Tags and receivers that track equipment location inside the hospital.
RV	Right ventricle	The chamber that pumps blood to the lungs.
SA	Sinoatrial node	The heart's natural pacemaker in the right atrium.
SaaS	Software as a service	Complete applications delivered over the web, such as email and office suites.
SAT	Source address table	The switch's table of which MAC address lives on which port.
SBOM	Software bill of materials	A list of the software components inside a device.
SBP	Systolic blood pressure	The peak arterial pressure during ventricular contraction.
SCD	Sequential compression device	Inflatable leg sleeves that inflate in sequence to prevent blood clots.
SCR	Silicon-controlled rectifier	A thyristor that switches large currents on when triggered and stays on until current stops.
SDS	Safety data sheet	The document describing a chemical's hazards, handling, and first aid.
SIMV	Synchronized intermittent mandatory ventilation	A ventilator mode that gives a set number of breaths synchronized with the patient's own efforts.
SMART	Self-Monitoring, Analysis, and Reporting Technology	Hard drive self-diagnostics that warn before failure.
SMB	Server Message Block	Windows file and printer sharing on TCP 445.
SMDA	Safe Medical Devices Act	The 1990 law requiring hospitals to report device-related deaths and serious injuries to the FDA and manufacturer.
SMPS	Switch-mode power supply	A power supply that switches at high frequency to regulate voltage efficiently.
SpO2	Peripheral oxygen saturation	The percentage of hemoglobin carrying oxygen, measured by a pulse oximeter.

ACRONYM	STANDS FOR	MEANING
SSD	Solid-state drive	Flash storage with no moving parts: fast, silent, and shock resistant.
SSH	Secure Shell	Encrypted remote command-line access and file transfer on TCP 22; replaces Telnet and FTP.
ST	ST segment	The ECG segment between the QRS and the T wave, where injury and ischemia show as elevation or depression.
STEL	Short-term exposure limit	The maximum average exposure over a 15-minute period.
SVT	Supraventricular tachycardia	A fast, regular rhythm originating above the ventricles, with narrow QRS.
TC	Transport Canada	The Canadian counterpart of the DOT for cylinder rules.
TCP	Transmission Control Protocol	Reliable, ordered, connection-based transport: three-way handshake, acknowledgments, retransmission.
TDR	Time-domain reflectometer	A cable tester that finds the distance to a break from signal reflections.
TENS	Transcutaneous electrical nerve stimulation	Low-level electrical pulses through skin electrodes for pain relief.
TLS	Transport Layer Security	The encryption layer under HTTPS and most secure protocols; use 1.2 or 1.3.
TLV	Threshold limit value	The ACGIH recommended exposure limit.
TOCO	Tocodynamometer	The pressure sensor on a fetal monitor that records uterine contractions.
TPN	Total parenteral nutrition	Complete nutrition delivered into a vein through an infusion pump.
TR	Telecommunications room	The wiring closet on each floor that serves nearby devices within the 90 m horizontal cable limit.
TV	Television	A display or broadcast receiver; smart TVs are IoT devices.
UDP	User Datagram Protocol	Connectionless, fast transport with no delivery guarantee; used for voice, video, DNS, DHCP.
UEFI	Unified Extensible Firmware Interface	Modern firmware that replaces BIOS: GPT disks, Secure Boot, mouse-driven setup.
UN	United Nations	The body whose Globally Harmonized System is the format hazard communication now uses.
UPS	Uninterruptible power supply	A battery unit that keeps equipment running through a power cut and conditions the power.
US	United States	The country; used for US standards, the 60 Hz mains frequency, and the AHA electrode color code.
USB	Universal Serial Bus	The standard port for peripherals, storage, and charging.
uV	Microvolt	One millionth of a volt; EEG signals are in microvolts.
UV	Ultraviolet	Light beyond violet, used for disinfection and in spectrophotometers, and absorbed by glass and plastic.
VA	Volt-ampere	The unit of apparent power, which a supply must deliver regardless of power factor.
VAR	Volt-ampere reactive	The unit of reactive power, which shuttles between source and load without doing work.
VF	Ventricular fibrillation	Chaotic ventricular activity with no output; the rhythm defibrillators treat.
VLAN	Virtual local area network	A logical LAN on a switch: ports in different VLANs are separate broadcast domains.
VM	Virtual machine	A complete computer running as software on a hypervisor.
VPN	Virtual private network	An encrypted tunnel across an untrusted network such as the internet.
VT	Ventricular tachycardia	A fast rhythm from the ventricles with wide QRS complexes; pulseless VT is shockable.
VT	Tidal volume	The volume of one breath, set on a ventilator at about 6 to 8 mL per kilogram of ideal body weight.
WAN	Wide area network	Connects LANs across cities or countries, usually over links leased from a provider.
WEP	Wired Equivalent Privacy	The original Wi-Fi encryption, broken in minutes; never use it.
Wi-Fi	Wireless fidelity	The trade name for IEEE 802.11 wireless networking.
WIPO	World Intellectual Property Organization	The UN agency for international copyright and patent treaties.
WLAN	Wireless local area network	A LAN delivered over Wi-Fi instead of cables.
WMTS	Wireless Medical Telemetry Service	The FCC-protected bands for patient telemetry: 608 to 614, 1395 to 1400, and 1427 to 1432 MHz.
WPA2	Wi-Fi Protected Access 2	Wi-Fi security using AES with CCMP; still common and acceptable.
WPA3	Wi-Fi Protected Access 3	The current Wi-Fi security standard: SAE in Personal mode, stronger encryption, protected management frames.
XML	Extensible Markup Language	A text format for structured data; HL7 version 3 and CDA use it.
YAG	Yttrium aluminum garnet (laser)	The crystal in Nd:YAG lasers, which emit near-infrared light that penetrates tissue.
YSI	Yellow Springs Instruments (temperature probe standard)	The company whose YSI 400 and YSI 700 thermistor curves became the standards for temperature probes.